

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460784.9348 +/- 0.0086 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.093 +/- 0.04
Transit depth (Rp/Rs)^2	0.86 +/- 0.74 [%]
Orbital Inclination (inc)	83.3 +/- 2.91
Airmass coefficient 1 (a1)	2.19 +/- 1.49
Airmass coefficient 2 (a2)	-0.61 +/- 0.8
Scatter in the residuals of the lightcurve fit is	69.02 %
Best Comparison Star	#1 - [349, 68]
Optimal Aperture	0.91
Optimal Annulus	7.3
Transit Duration (day)	0.036 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.093 $\pm$ 0.04 vs 0.119 (deviation 0.65 σ)
Duration fit vs published	0.036 d vs 0.07 d (deviation 0.89 σ)
Mid-time O-C	17.6 min
Depth SNR	2.3
Residual scatter	69.02 %

Light curve

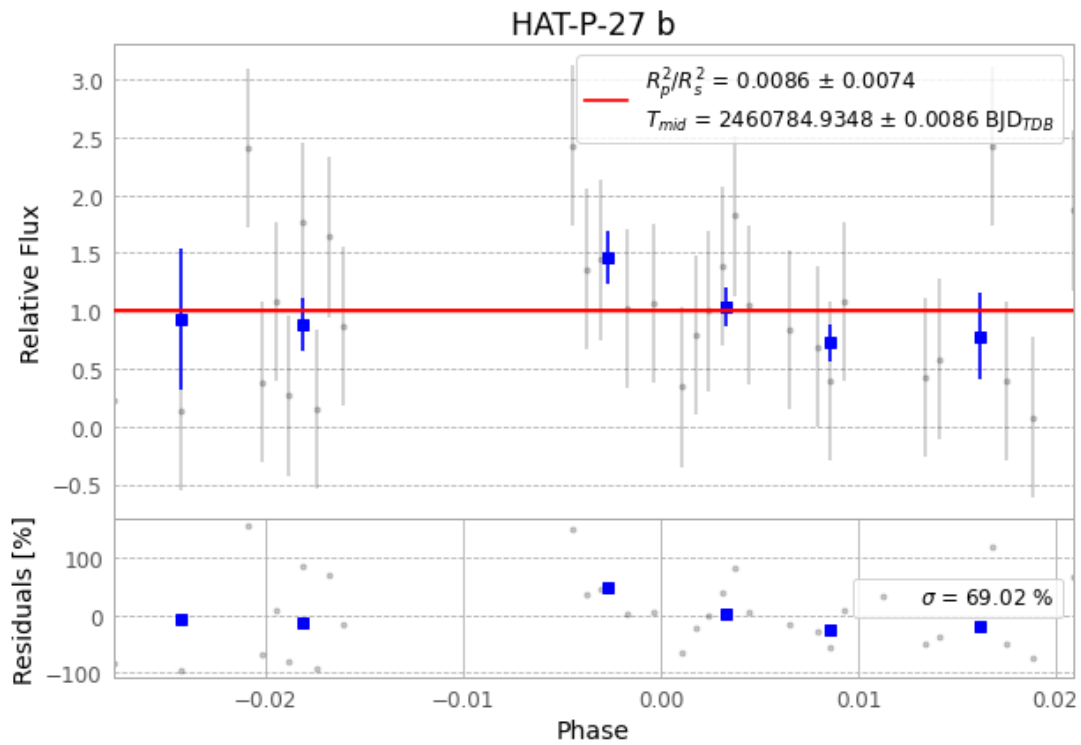


Figure 1 — FinalLightCurve_HAT-P-27 b_2025-04-19.png (SHA-256 974d14a0c351d123...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [444, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ac76ab90d594e4e4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

73 FITS frames (48 MB), HATP-27250419081516.FITS → HATP-27250419115115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-250419.log (e154e6472bea...), FinalLightCurve_HAT-P-27 b_2025-04-19.png (974d14a0c351...), FinalLightCurve_HAT-P-27 b_2025-04-19.csv (8c934c992a84...)

Compute resources

Wall time 185.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:41Z.